

Swaham Pitambar Patra

Duck Typing:

```
1  class Duck:
2      def sound(self):
3          print("quack")
4
5  class AnotherBird:
6      def sound(self):
7          print("Bird similar to duck")
8
9  def makeSound(bird):
10     bird.sound()
11
12 # Creating Instances
13 duck = Duck()
14 anotherBird = AnotherBird()
15
16 # Calling Methods
17 makeSound(duck)
18 makeSound(anotherBird)
```

PROBLEMS TERMINAL DEBUG CONSOLE PORTS

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Use
quack
Bird similar to duck
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>
```

Method Overriding:

```
1 # Parent Class
2 class Parent:
3     def myMethod(self):
4         print("This is parent class.")
5
6 # Child Class
7 class Child(Parent):
8     def myMethod(self):
9         print("This is Parent Class method in Child Class(Overridden)")
10
11 # Instance of child
12
13 c = Child()
14
15 c.myMethod()
```

PROBLEMS TERMINAL DEBUG CONSOLE PORTS

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python+DS\Tasks 09-07\
This is Parent Class method in Child Class(Overridden)
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>
```

Method Overloading:

```
1 class example:
2     def add(self, a,b):
3         x = a+b
4         return x
5
6     def add(self, a,b,c):
7         x = a+b+c
8         return x
9
10 obj = example()
11 print(f"{obj.add(10,20)}")
12 print(f"{obj.add(1,2,2)}")
```

PROBLEMS TERMINAL DEBUG CONSOLE PORTS

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python+DS\Tasks 09-07\polymo
Traceback (most recent call last):
  File "c:\Users\Swaham Patra\OneDrive\ITR Python+DS\Tasks 09-07\polymorphism\method_overloading.py", line 11, in <module>
    print(f"{obj.add(10,20)}")
    ~~~~~~^~~~~~
TypeError: example.add() missing 1 required positional argument: 'c'
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>
```

Operator Overloading:

```
1 print(1+2)
2 print("Geeks"+" for")
3 print(3*4)
4 print("Geeks,"*4)
```

PROBLEMS TERMINAL DEBUG CONSOLE PORTS

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>
3
Geeks for
12
Geeks,Geeks,Geeks,Geeks,
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>
```

Abstraction:

```
1 from abc import ABC, abstractmethod # abc = Abstract Base Class
2
3 class Shape:
4     @abstractmethod
5     def area(self):
6         pass
7
8 class Square(Shape):
9     def __init__(self,side):
10        self.side = side
11    def area(self):
12        return self.side * self.side
13
14 S1 = Square(10)
15 print(S1.area())
```

PROBLEMS TERMINAL DEBUG CONSOLE PORTS

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python+DS\Tasks C
100
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>
```